

Case Study No.: 6

1. Case Study Title: To demonstrate JavaScript string operations using `substring()`, `indexOf()`, `split()`, `replace()`, string reversal, and vowel counting based on user input.

2. Case Study Program Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>String Operations</title>

  <style>
    body {
      background-color: rgb(239, 171, 243);
      font-family: Arial, sans-serif;
      padding: 20px;
    }
  </style>
</head>
<body>
```

```
}
```

```
input {  
    padding: 8px;  
    margin: 5px 0 15px 0;  
    width: 350px;  
    display: block;  
}
```

```
textarea {  
    padding: 10px;  
    margin: 5px 0 15px 0;  
    width: 500px;  
    height: 80px;  
    display: block;
```

```
    }

    button {
        padding: 10px 20px;
        margin: 10px 0;
        cursor: pointer;
    }

    #output {
        margin-top: 20px;
        font-size: 18px;
        line-height: 1.6;
    }
</style>
</head>

<body>

    <h2>Sonali's JavaScript String Operations</h2>

    <label for="str">Enter String:</label>
        <textarea id="str" placeholder="Enter your string or
paragraph"></textarea>

    <label for="start">Substring Start Index:</label>
    <input type="number" id="start" placeholder="Enter start index (e.g.,
0)">

    <label for="end">Substring End Index:</label>
    <input type="number" id="end" placeholder="Enter end index">

    <label for="search">IndexOf - Enter text to search:</label>
    <input type="text" id="search" placeholder="Enter character or word">
```

```

        // Substring (Handles cases where end index isn't specified)
        let subResult = (end > 0) ? str.substring(start, end) :
str.substring(start);
        output += "<strong>Substring:</strong> " + subResult + "<br>";

        // indexOf
        if (search) {
            output += "<strong>Index of '" + search + "':</strong> " +
str.indexOf(search) + "<br>";
        }

        // Split
        if (separator) {
            let sepChar = (separator === "space") ? " " : separator;
            output += "<strong>Split:</strong> [" + str.split(sepChar).join(",
") + "]<br>";
        }

        // Replace (Replaces all instances safely)
        if (oldText) {
            output += "<strong>Replace:</strong> " + str.replaceAll(oldText,
newText) + "<br>";
        }

        // Reverse
        let reverse = str.split("").reverse().join("");
        output += "<strong>Reversed String:</strong> " + reverse + "<br>";

        // Count Vowels (Using regex for cleaner execution)
        let vowelMatch = str.match(/[aeiou]/gi);
        let count = vowelMatch ? vowelMatch.length : 0;

        output += "<strong>Number of Vowels:</strong> " + count;

        document.getElementById("output").innerHTML = output;
    }
</script>
</body>
</html>

```